

BUSINESS REQUIREMENTS DOCUMENT

Meta Ad Performance Analysis

Facebook & Instagram Campaign Analytics Dashboard

PROJECT OVERVIEW

A structured requirements document for designing a Power BI report that tracks campaign reach, engagement, conversions, audience response, and budget utilization across Meta advertising platform.

1. Executive Summary

The business requires a performance tracking report for paid advertising campaigns running on Facebook and Instagram. The report will provide a unified view of campaign reach, audience engagement, conversions, and budget utilization so that the marketing team can evaluate performance and make better allocation decisions.

Primary outcome: Build an interactive Power BI dashboard that enables dynamic analysis by platform, audience segment, geography, time, and ad type.

2. Business Objectives

- Identify the most effective advertising platform: Facebook or Instagram.
- Track campaign return on investment and support budget optimization.
- Understand audience engagement patterns across gender, age group, country, month, week, and hour.
- Compare campaign and ad-format performance using consistent, clearly defined KPIs.

3.KPI Framework

The following KPI definitions provide a consistent measurement framework for the dashboard. Event-based metrics are derived from `ad_events[event_type]`, while budget metrics are sourced from the `campaigns` table.

KPI	Definition	Formula / Logic	Business Use
Impressions	Number of times ads were displayed.	Count of event_type = Impression	Measure reach
Clicks	Number of times users clicked ads.	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared.	Count of event_type = Share	Measure viral engagement
Comments	Number of user comments on ads.	Count of event_type = Comment	Assess sentiment and feedback
Purchases	Number of purchases made after seeing ads.	Count of event_type = Purchase	Measure conversions
Engagements	Total interactions generated by an ad.	Clicks + Shares + Comments	Measure engagement volume
CTR	Percentage of impressions that resulted in clicks.	$(\text{Clicks} \div \text{Impressions}) \times 100$	Assess ad effectiveness
Engagement Rate	Percentage of impressions that resulted in engagements.	$(\text{Engagements} \div \text{Impressions}) \times 100$	Assess overall ad appeal
Conversion Rate	Percentage of clicks that resulted in purchases.	$(\text{Purchases} \div \text{Clicks}) \times 100$	Measure funnel efficiency
Purchase Rate	Percentage of impressions that resulted in purchases.	$(\text{Purchases} \div \text{Impressions}) \times 100$	Measure conversion from reach
Total Budget	Total spend allocated to campaigns.	<code>SUM(campaigns[total_budget])</code>	Support cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign.	$\text{Total Budget} \div \text{Campaign Count}$	Evaluate budget distribution

4. Dashboard Visualization Requirements

The dashboard should use a dynamic metric selector so users can switch the analysis between key measures such as Impressions, Clicks, Purchases, and Engagements.

01 Target Gender — Donut Chart

Requirement	Visualize performance split by target gender from the ads table. The selected metric should change dynamically through a parameter.
Purpose	Identify which gender segment contributes most to the selected metric.

02 Target Age Group — Bar Chart

Requirement	Display engagement across age groups defined in the ads table. Each bar represents one age group and the metric switches dynamically.
Purpose	Highlight the age group most responsive to campaigns.

03 Country — Map

Requirement	Display performance by country using the users table. Bubble size or color intensity should represent the selected metric.
Purpose	Provide a geographic view of campaign reach and engagement.

04 Calendar Month — Calendar Heat Map

Requirement	Plot performance at the monthly level using ad_events[timestamp]. Darker shades indicate higher activity.
Purpose	Detect seasonal trends, peak ad months, and low-activity periods.

05 Weekly Trend — Stacked Column Chart

Requirement	Show weekly performance with week number on the X-axis, ad_type as the stack, and the selected metric on the Y-axis.
Purpose	Compare ad-type contribution across weeks.

06 Hourly Trend — Area Chart

Requirement	Show activity by hour using ad_events[time_of_day], with hours 0–23 on the X-axis and the selected metric on the Y-axis.
Purpose	Understand user activity patterns throughout the day.

07 Ad Type — Matrix

Requirement	Show the selected metric across ad types, with optional platform columns for Facebook and Instagram.
Purpose	Compare ad formats and platforms side by side.

5. Expected Business Value

- A single source of truth for paid Meta campaign performance.
- Faster identification of high-performing platforms, audiences, ad types, locations, and time periods.
- Clearer visibility into the relationship between reach, engagement, conversion, and budget.
- Improved decision-making for campaign optimization and future budget allocation.

6. Deliverable Summary

Deliverable	Description
Power BI Dashboard	Interactive performance analysis for Facebook and Instagram paid campaigns.
Dynamic KPI Selector	User-controlled switching between campaign performance measures.
Audience Analysis	Gender, age group, and country performance breakdowns.
Time Analysis	Monthly, weekly, and hourly performance trends.
Ad Format Comparison	Matrix and trend views comparing ad types and platforms.